

Digital Principles and Computer Organisation (DPCO)

INTRODUCTION TO K-MAPS

- Visual tool used to simplify Boolean algebra expressions
- Maurice Karnaugh introduced the technique in 1953 as a refinement of Edward W. Veitch's 1952 Veitch chart

INTRODUCTION TO K-MAPS

- Systematic method to obtain **simplified (minimal) sum-of-products (SOP) expressions**.
- Objective: *Fewest* possible product terms and literals.
- Diagrammatic technique based on a special form of *Venn diagram*.
- Advantage: Easy to use.
- Disadvantage: Limited to 5 or 6 variables.

2-VARIABLE K-MAPS (1 / 4)

- Karnaugh-map (K-map) is an abstract form of Venn diagram, organised as **a matrix of squares**, where
 - Each square represents a **minterm**
 - Two adjacent squares represent minterms that **differ by exactly one literal**
- Let the 2 variables be a and b. The K-map can be drawn as...

2-VARIABLE K-MAPS (2/4)

- Alternative **layouts** of a 2-variable (a, b) K-map:

Alternative 1:

	b	
	a'b'	a'b
a	ab'	ab

O
R

	b	
	<i>m0</i>	<i>m1</i>
a	<i>m2</i>	<i>m3</i>

Alternative 2:

	a	
	a'b'	ab'
b	a'b	ab

O
R

	a	
	<i>m0</i>	<i>m2</i>
b	<i>m1</i>	<i>m3</i>

Alternative 3:

	a	
	ab	a'b
b	ab'	a'b'

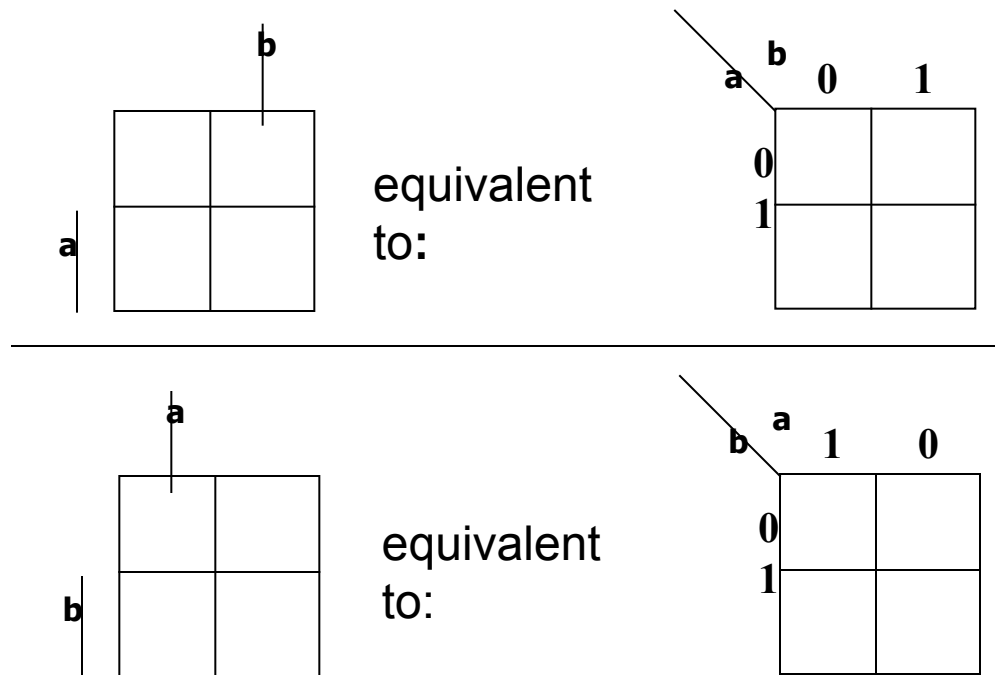
O
R

	a	
	<i>m3</i>	<i>m1</i>
b	<i>m2</i>	<i>m0</i>

and others...

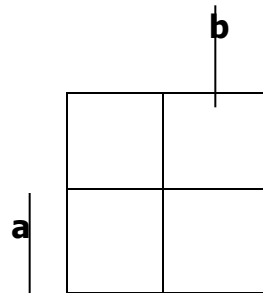
2-VARIABLE K-MAPS (3/4)

- Alternative **labelling** of a 2-variable (a, b) K-map:

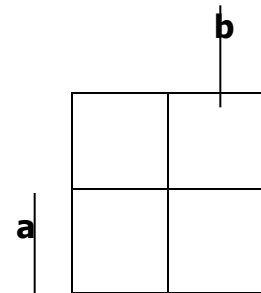


2-VARIABLE K-MAPS (4/4)

- The K-map for a function is filled by putting
 - A '1' in the square the corresponds to a minterm of the function
 - A '0' otherwise
- Example: Half adder.



$$C = a \cdot b$$



$$S = a \cdot b' + a' \cdot b$$

3-VARIABLE K-MAPS (1/2)

- As there are 8 minterms for 3 variables, so there are 8 squares in a 3-variable K-map.
- Example: Let the variables be a , b , c .

Example: Let the variables be a, b, c and the function be $f(a, b, c) = a'b'c + a'bc' + abc$

O
R

a, b, c.

Diagram illustrating a Karnaugh map for variables a, b, and c. The map is a 2x4 grid with columns labeled 00, 01, 11, 10 (representing bc) and rows labeled 0, 1 (representing a). The cells contain minterms m0, m1, m3, m2 in the top row and m4, m5, m7, m6 in the bottom row. A green oval highlights the top row (m0, m1, m3, m2). A blue box at the bottom contains the text "Note Gray code sequence".

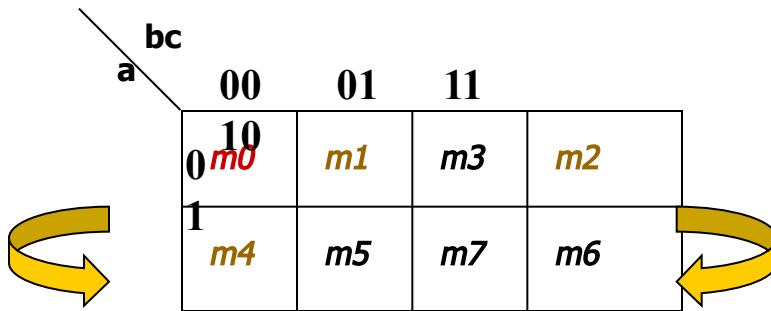
	00	01	11	10
0	m0	m1	m3	m2
1	m4	m5	m7	m6

Note Gray code sequence

Above arrangement ensures that minterms of adjacent cells **differ by only ONE literal**.
(Other arrangements which satisfy this criterion may also be used.)

3-VARIABLE K-MAPS (2/2)

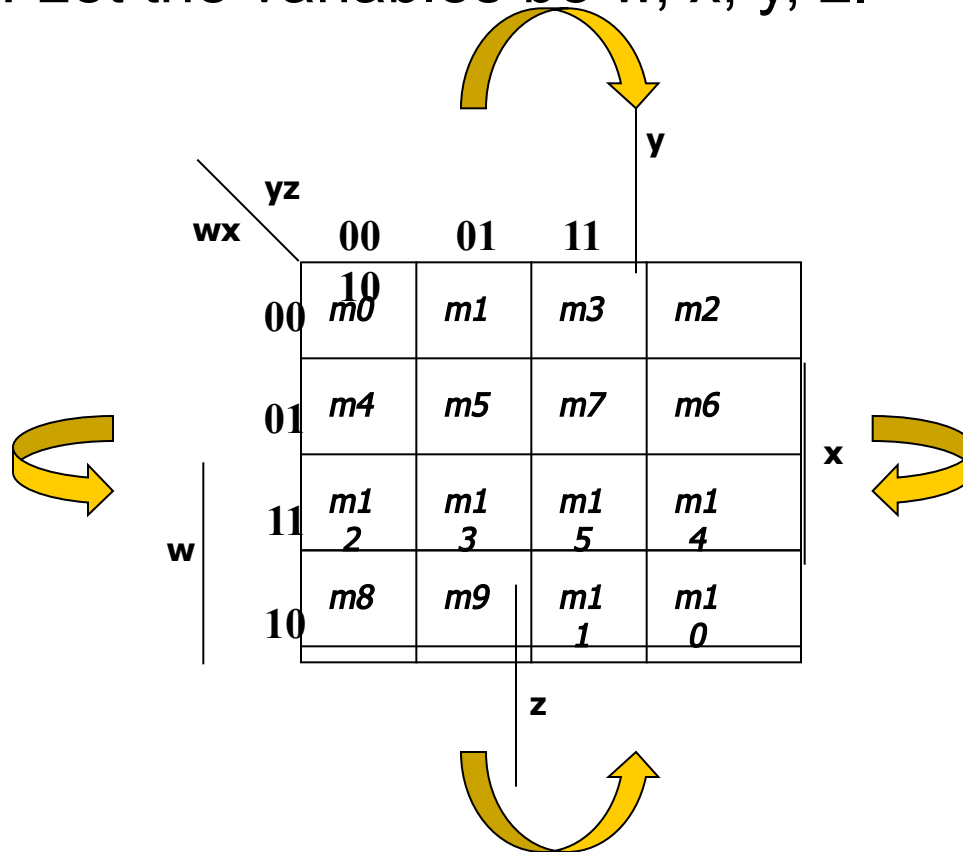
- There is **wrap-around** in the K-map:
 - $a' \cdot b' \cdot c'$ (m_0) is adjacent to $a' \cdot b \cdot c'$ (m_2)
 - $a \cdot b' \cdot c'$ (m_4) is adjacent to $a \cdot b \cdot c'$ (m_6)



Each cell in a 3-variable K-map has 3 adjacent neighbours. In general, each cell in an n -variable K-map has n adjacent neighbours. For example, m_0 has 3 adjacent neighbours: m_1 , m_2 and m_4 .

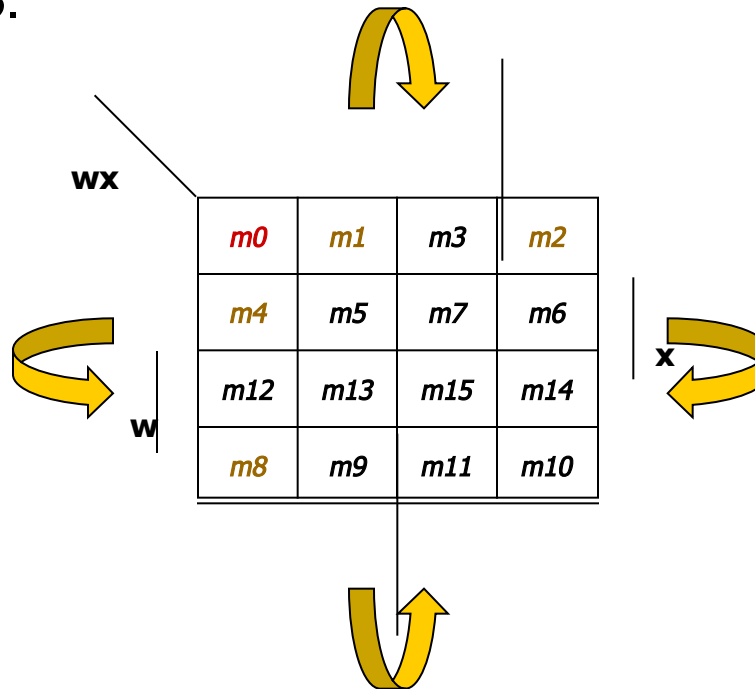
4-VARIABLE K-MAPS (1/2)

- There are 16 square cells in a 4-variable K-map.
- Example: Let the variables be w, x, y, z .



4-VARIABLE K-MAPS (2/2)

- There are 2 wrap-arounds.
- Every cell has 4 neighbours.
 - Example: The cell corresponding to minterm $m0$ has neighbours $m1$, $m2$, $m4$ and $m8$.



HOW TO USE K-MAPS (1/7)

- Based on the **Unifying Theorem** (complement law):

$$\mathbf{A + A' = 1}$$

- In a K-map, each cell containing a '1' corresponds to a minterm of a given function F .
- Each valid grouping of adjacent cells containing '1' then corresponds to a **simpler product term** of F .
 - A group must have size in **powers of two**: 1, 2, 4, 8, ...
 - Grouping 2 adjacent cells eliminates 1 variable from the product term; grouping 4 cells eliminates 2 variables; grouping 8 cells eliminates 3 variables, and so on. In general, grouping 2^n cells eliminates n variables.

HOW TO USE K-MAPS (2/7)

- **Group as many cells as possible**
 - The larger the group, the fewer the number of literals in the resulting product term.
- **Select as few groups as possible to cover all the cells (minterms) of the function**
 - The fewer the groups, the fewer is the number of product terms in the simplified (minimal) SOP expression.

HOW TO USE K-MAPS (3/7)

- Example:

$$\begin{aligned} F(w,x,y,z) &= w' \cdot x \cdot y' \cdot z' + w' \cdot x \cdot y' \cdot z + w \cdot x' \cdot y \cdot z' \\ &\quad + w \cdot x' \cdot y \cdot z + w \cdot x \cdot y \cdot z' + w \cdot x \cdot y \cdot z \\ &= \Sigma m(4, 5, 10, 11, 14, 15) \end{aligned}$$

		yz			x
		00	01	11	
wx	00	10			
	01	1	1		
	11			1	1
	10			1	1

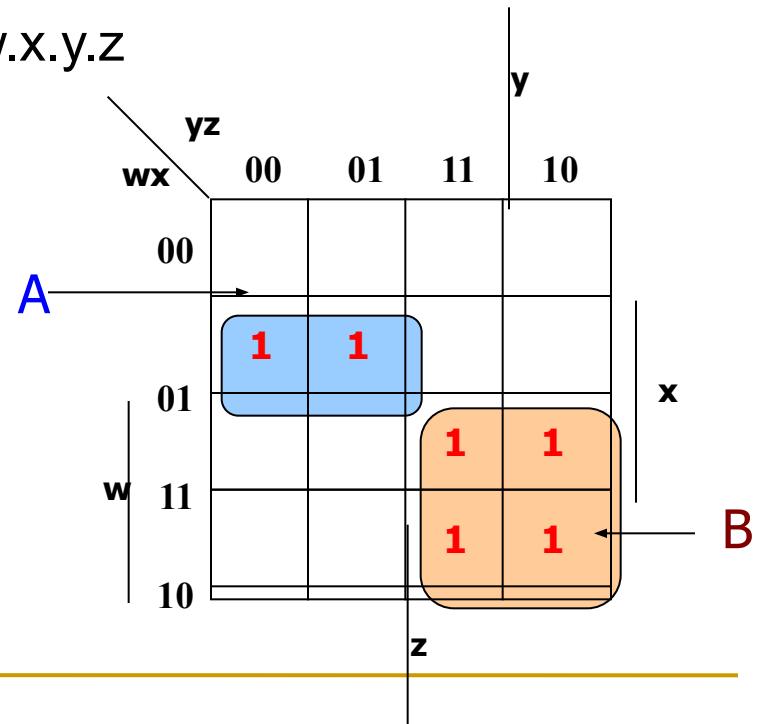
(cells with '0' are not shown for clarity)

HOW TO USE K-MAPS (4/7)

- Each group of adjacent minterms corresponds to a possible product term of the given function.
- Here, there are 2 groups of minterms, A and B:

$$A = w'.x.y'.z' + w'.x.y'.z = w'.x.y'.(z' + z) = \mathbf{w'.x.y'}$$

$$\begin{aligned} B &= w.x'.y.z' + w.x'.y.z + w.x.y.z' + w.x.y.z \\ &= w.x'.y.(z' + z) + w.x.y.(z' + z) \\ &= w.x'.y + w.x.y \\ &= w.(x' + x).y \\ &= \mathbf{w.y} \end{aligned}$$



HOW TO USE K-MAPS (5/7)

- Each product term that corresponds to a group, $w' \cdot x \cdot y'$ and $w \cdot y$, represents the sum of minterms in that group.
- Boolean expression is therefore the sum of product terms (SOP) that represent all groups of the minterms of the function:

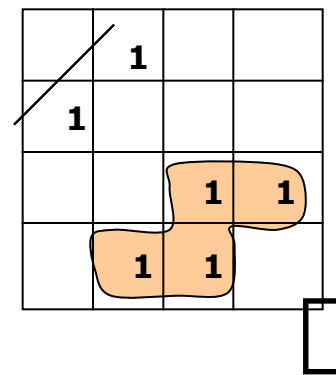
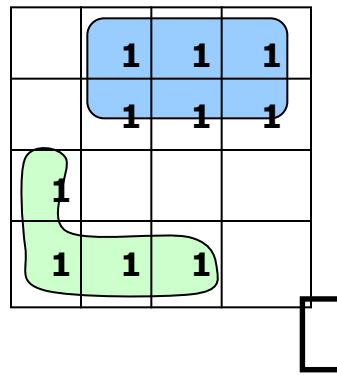
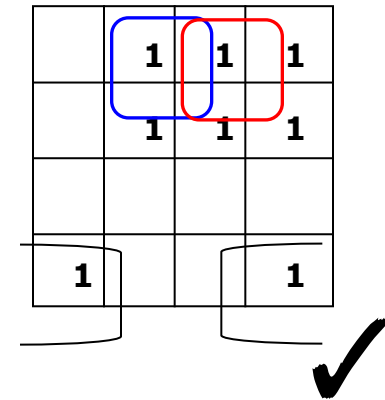
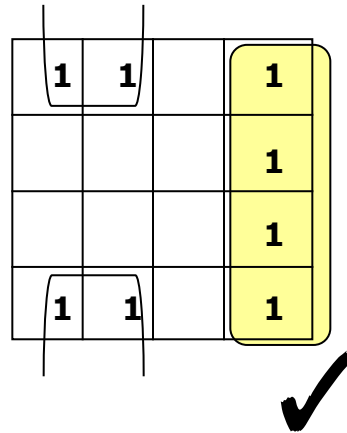
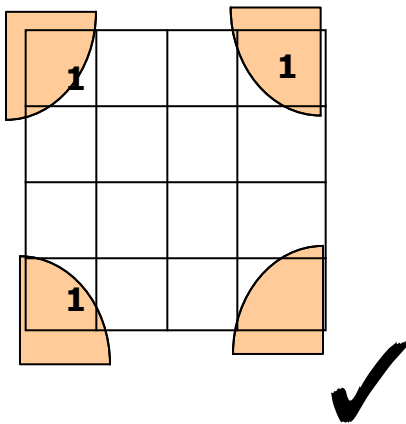
$$F(w,x,y,z) = \text{group A} + \text{group B} = w' \cdot x \cdot y' + w \cdot y$$

HOW TO USE K-MAPS (6/7)

- The larger the group (the more minterms it contains), the fewer is the number of literals in the associated product term.
 - Recall that a group must have size in powers of two.
 - Example: For a 4-variable K-map with variables w, x, y, z
 - 1 cell = 4 literals. Examples: $w \cdot x \cdot y \cdot z$, $w' \cdot x \cdot y' \cdot z$
 - 2 cells = 3 literals. Examples: $w \cdot x \cdot y$, $w \cdot y' \cdot z'$
 - 4 cells = 2 literals. Examples: $w \cdot x$, $x' \cdot y$
 - 8 cells = 1 literal. Examples: w , y' , z
 - 16 cells = no literal (i.e. logical constant 1). Example: 1

HOW TO USE K-MAPS (7/7)

- Examples of valid and invalid groupings.



CONVERTING TO MINTERMS FORM (1/2)

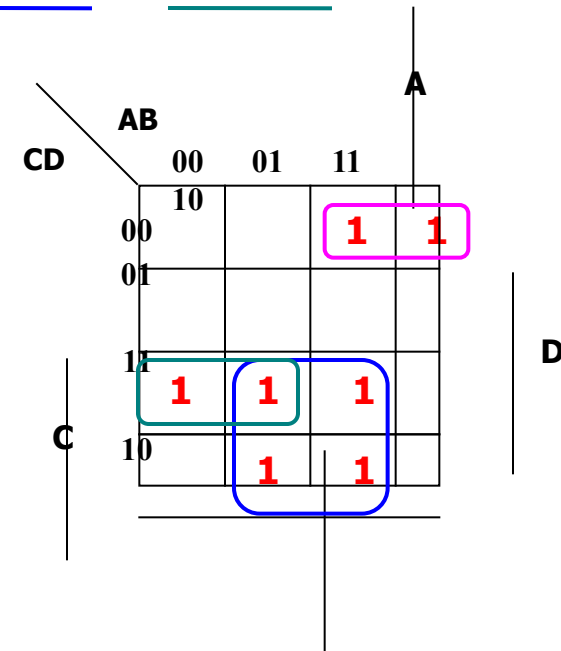
- The K-map of a function can be easily filled in when the function is given in sum-of-minterms form.
- What if it is not in sum-of-minterms form?
 - Convert it into sum-of-products (SOP) form
 - Expand the SOP expression into sum-of-minterms expression, or fill in the K-map directly based on the SOP expression.

CONVERTING TO MINTERMS FORM (2/2)

■ Example:

$$\begin{aligned}
 F(A,B,C,D) &= A.(C+D)'.(B'+D') + C.(B+C'+A'.D) \\
 &= A.(C'.D').(B'+D') + B.C + C.C' + A'.C.D \\
 &= A.B'.C'.D' + A.C'.D' + B.C + A'.C.D
 \end{aligned}$$

$$\begin{aligned}
 &A.B'.C'.D' + A.C'.D' + B.C + A'.C.D \\
 &= A.B'.C'.D' + A.C'.D'.(B+B') + B.C + A'.C.D \\
 &= \underline{A.B'.C'.D'} + A.B.C'.D' + \underline{A.B'.C'.D'} + B.C.(A+A') \\
 &\quad + A'.C.D \\
 &= A.B'.C'.D' + A.B.C'.D' + A.B.C + A'.B.C + \\
 &\quad A'.C.D \\
 &= A.B'.C'.D' + A.B.C'.D' + A.B.C.(D+D') + \\
 &\quad A'.B.C.(D+D') + A'.C.D.(B+B') \\
 &= A.B'.C'.D' + A.B.C'.D' + A.B.C.D + A.B.C.D' + \\
 &\quad A'.B.C.D + A'.B.C.D' + A'.B'.C.D
 \end{aligned}$$

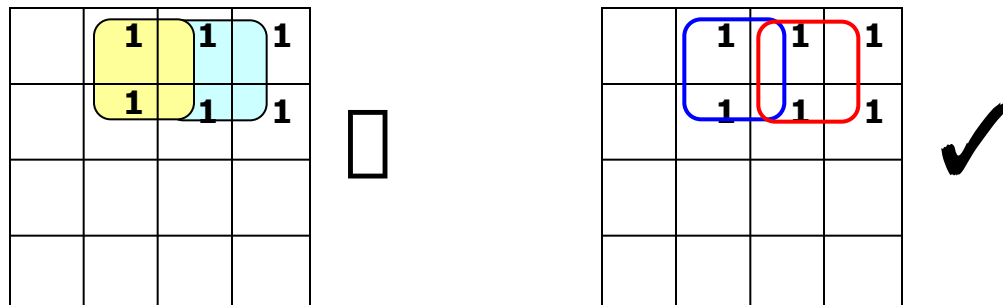


PIs AND EPIs (1 / 3)

- To find the simplest (minimal) SOP expression from a K-map, you need to obtain:
 - Minimum number of literals per product term; and
 - Minimum number of product terms.
- Achieved through K-map using
 - *Bigger groupings* of minterms (**prime implicants**) where possible; and
 - *No redundant groupings* (look for **essential prime implicants**)
- **Implicant**: a product term that could be used to cover minterms of the function.

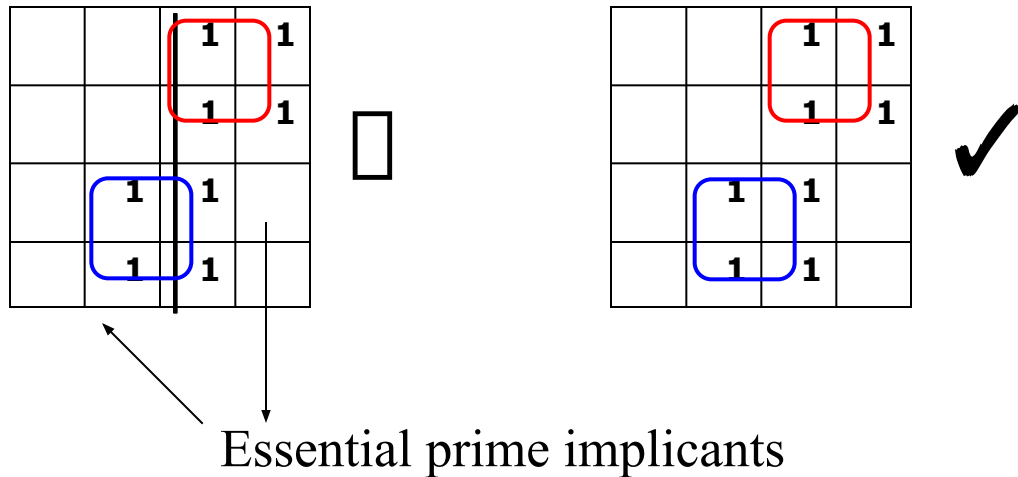
PIs AND EPIs (2/3)

- **Prime implicant (PI)**: a product term obtained by combining the *maximum possible number of minterms* from *adjacent* squares in the map. (That is, it is the biggest grouping possible.)
- Always look for prime implicants in a K-map.



PIs AND EPIs (3/3)

- No redundant groups:



- **Essential prime implicant (EPI):** a prime implicant that includes at least one minterm that is not covered by any other prime implicant.

QUICK REVIEW QUESTIONS (2)

- DLD pages 104, question 5-3.

5-3. Identify the prime implicants and essential prime implicants of the two K-maps below.

		bc			
		00	01	11	10
a	0	1	1	0	1
	1	0	1	0	0

		AB			A
		00	01	11	
CD	00	1	1		1
	01			1	1
	11	1	1		1
	10	1		1	1



EXAMPLE ON FINDING MINIMAL SOP EXPRESSION (1/4)

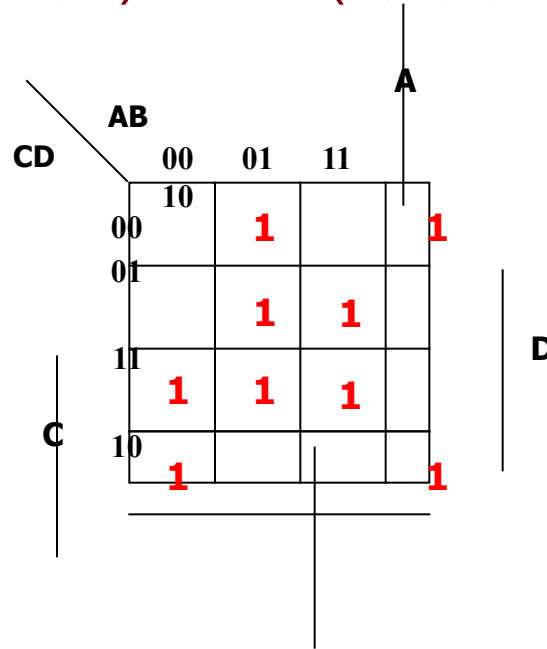
■ Algorithm

1. Circle all prime implicants on the K-map.
2. Identify and select all essential prime implicants for the cover.
3. Select a minimum subset of the remaining prime implicants to complete the cover, that is, to cover those minterms not covered by the essential prime implicants.

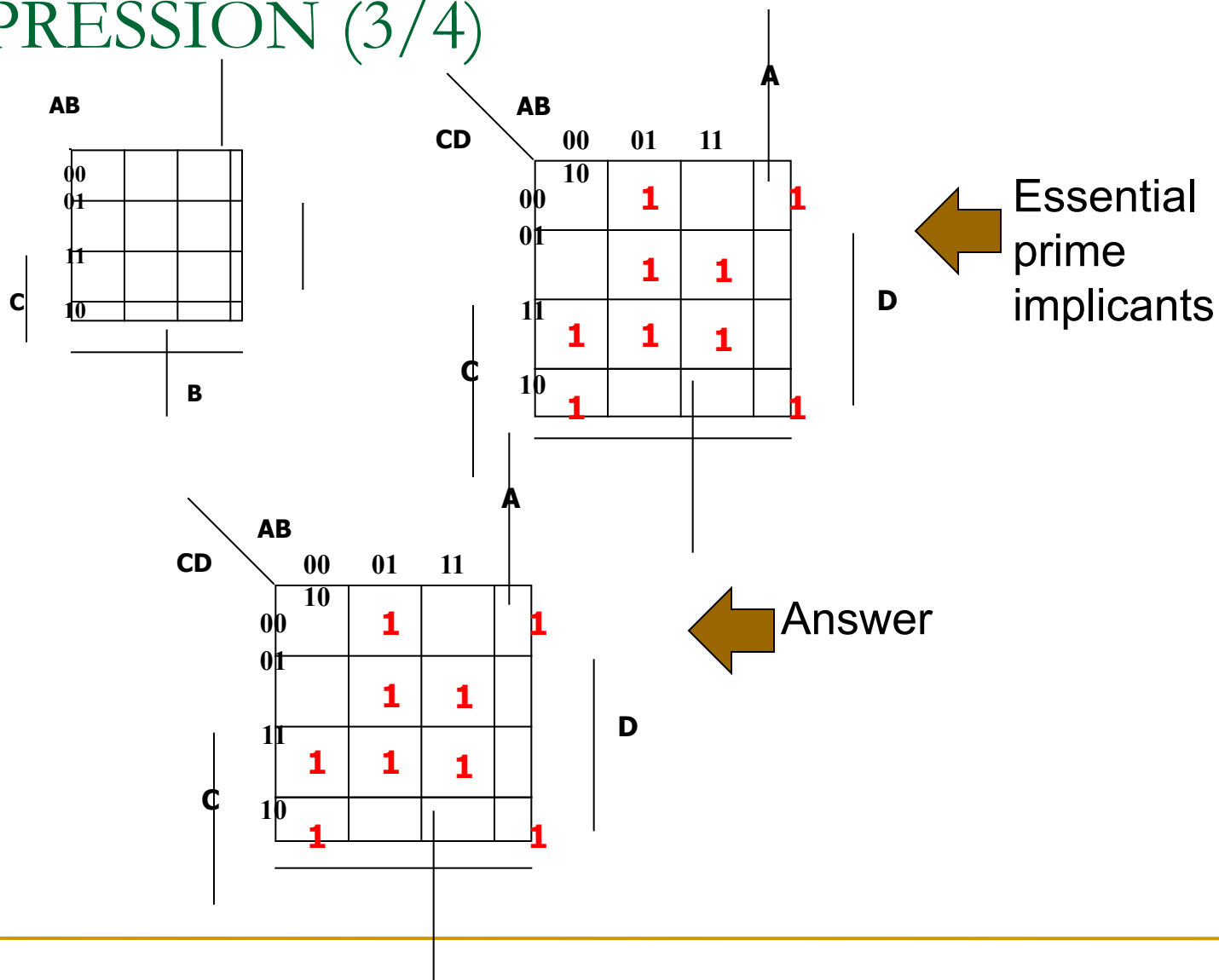
EXAMPLE ON FINDING MINIMAL SOP EXPRESSION (2/4)

■ Example #1:

$$F(A,B,C,D) = \sum m(2,3,4,5,7,8,10,13,15)$$



EXAMPLE ON FINDING MINIMAL SOP EXPRESSION (3/4)



EXAMPLE ON FINDING MINIMAL SOP EXPRESSION (4/4)

A 4x4 Karnaugh Map for the function F(A,B,C,D). The horizontal axis is labeled 'AB' with values 00, 01, 11. The vertical axis is labeled 'CD' with values 00, 01, 11, 10. The map contains red '1's in the following cells: (AB=00, CD=00), (AB=01, CD=00), (AB=01, CD=01), (AB=11, CD=01), (AB=00, CD=11), (AB=01, CD=11), (AB=11, CD=11), (AB=00, CD=10), and (AB=10, CD=10). The map is surrounded by labels A, B, C, and D with arrows indicating their positions.

AB \ CD	00	01	11
00	1	1	
01		1	1
11	1	1	1
10	1		

$F(A,B,C,D) =$

QUICK REVIEW QUESTIONS (3)

- DLD pages 104-105, questions 5-4 to 5-7.

5-4. Find the minimal SOP expression for $G(A,B,C,D)$.

CD \ AB		00	01	11	A
		10			
00	00		1		
	01		1	1	
11		1	1	1	D
10				1	

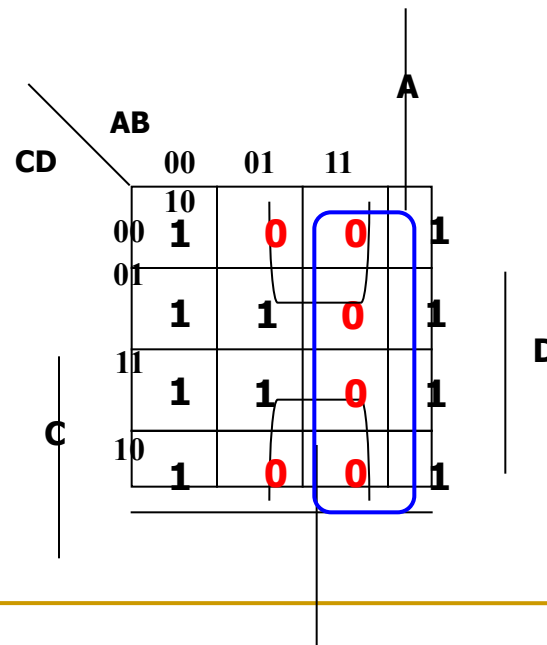


FINDING POS EXPRESSION (1/2)

- **Simplified POS expression** can be obtained by grouping the maxterms (i.e. 0s) of the given function.

- **Example:**

Given $F = \Sigma m(0,1,2,3,5,7,8,9,10,11)$, we first draw the K-map, then group the maxterms together:



FINDING POS EXPRESSION (2/2)

K-map of **F**

		AB			
		00	01	11	
CD	00	1	0	0	1
	01	1	1	0	1
C	11	1	1	0	1
	10	1	0	0	1

D

K-map of **F'**

		AB			
		00	01	11	
CD	00	0	1	1	0
	01	0	0	1	0
C	11	0	0	1	0
	10	0	1	1	0

D

- This gives the SOP of F' to be

$$F' = B \cdot D' + A \cdot B$$
- To get POS of F, we have

$$F =$$

DON'T-CARE CONDITIONS (1/3)

- In certain problems, some outputs are not specified or are invalid.
- Hence, these outputs can be either '1' or '0'.
- They are called **don't-care conditions**, denoted by X (or sometimes, d).
- Example: An odd parity generator for BCD code which has 6 unused combinations.

A	B	C	D	P
0	0	0	0	1
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0
1	0	0	0	0
1	0	0	1	1
1	0	1	0	X
1	0	1	1	X
1	1	0	0	X
1	1	0	1	X
1	1	1	0	X
1	1	1	1	X

DON'T-CARE CONDITIONS (2/3)

- Don't-care conditions can be used to help simplify Boolean expression further in K-maps.
- They could be chosen to be either '1' or '0', depending on which choice results in a simpler expression.
- We usually use the notation Σd to denote the set of don't-care minterms.
 - For example, the function P in the odd-parity generator for BCD code can be written as:

$$P = \Sigma m(0, 3, 5, 6, 9) + \Sigma d(10, 11, 12, 13, 14, 15)$$

DON'T-CARE CONDITIONS (3/3)

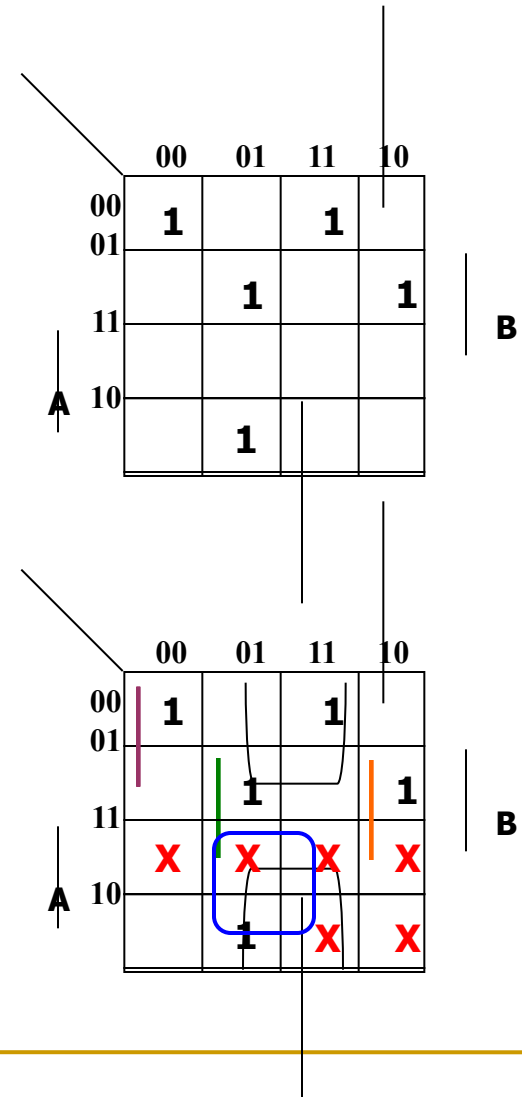
■ Comparison

- Without don't-cares:

$$P = A' \cdot B' \cdot C' \cdot D' + A' \cdot B' \cdot C \cdot D + A' \cdot B \cdot C' \cdot D + A' \cdot B \cdot C \cdot D' + A \cdot B' \cdot C' \cdot D$$

- With don't-cares:

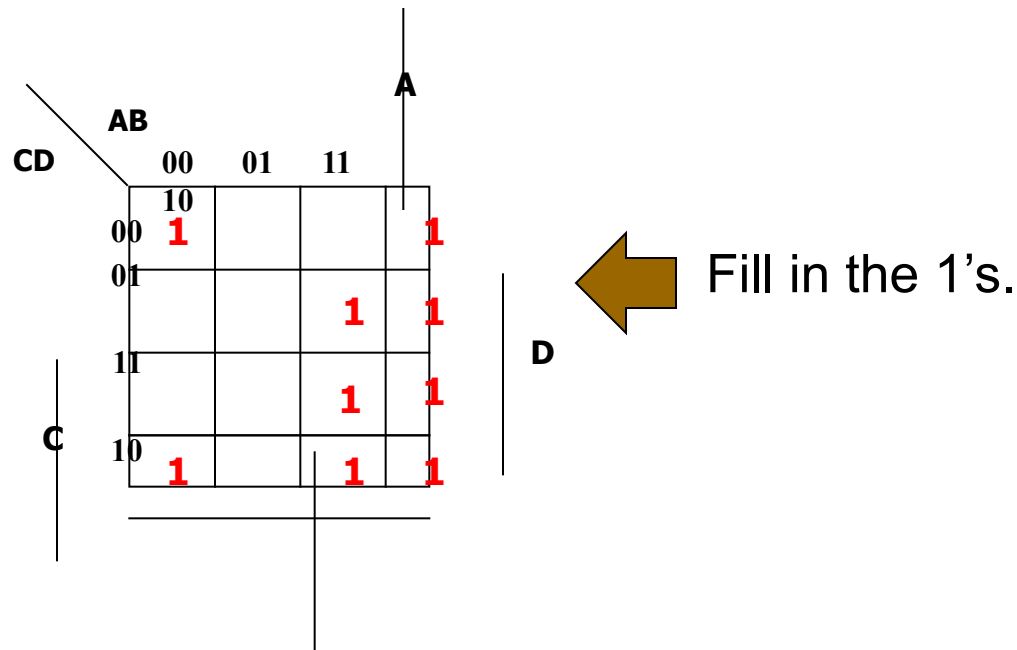
$$P = A' \cdot B' \cdot C' \cdot D' + B' \cdot C \cdot D + B \cdot C' \cdot D + B \cdot C \cdot D' + A \cdot D$$



MORE EXAMPLES (1/6)

- Example #2:

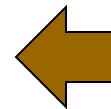
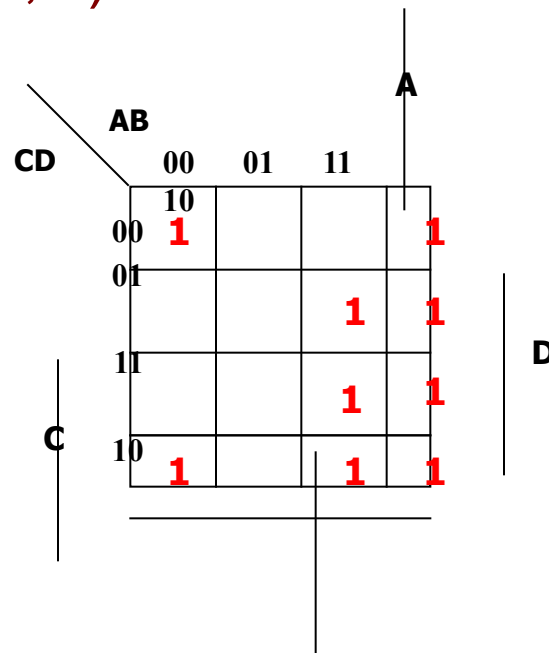
$$F(A,B,C,D) = A \cdot B \cdot C + B' \cdot C \cdot D' + A \cdot D + B' \cdot C' \cdot D'$$



MORE EXAMPLES (2/6)

■ Example #2:

$$F(A,B,C,D) = A \cdot B \cdot C + B' \cdot C \cdot D' + A \cdot D + B' \cdot C' \cdot D'$$



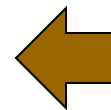
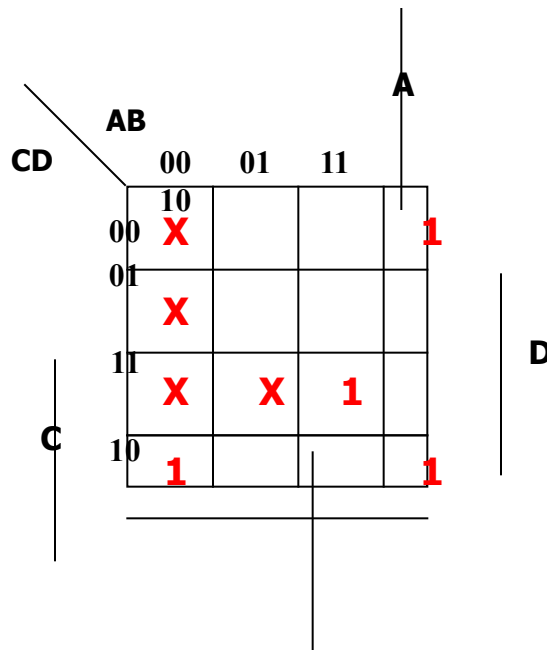
Find all PIs:

So the answer is: $F(A,B,C,D) =$

MORE EXAMPLES (3/6)

- Example #3 (with don't-cares):

$$F(A,B,C,D) = \Sigma m(2,8,10,15) + \Sigma d(0,1,3,7)$$

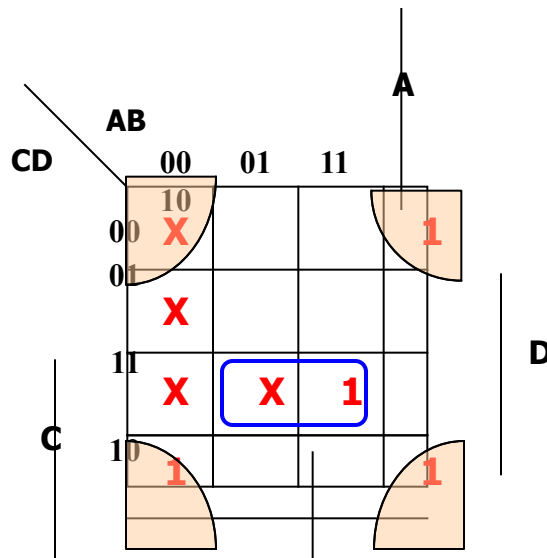


Fill in the 1's and X's.

MORE EXAMPLES (4/6)

- Example #3 (with don't-cares):

$$F(A,B,C,D) = \Sigma m(2,8,10,15) + \Sigma d(0,1,3,7)$$



Do we need to have an additional term $A'B'$ to cover the 2 remaining X's?

Answer: $F(A,B,C,D) =$

MORE EXAMPLES (5/6)

- Find the simplest POS expression for example #2:
$$F(A,B,C,D) = A \cdot B \cdot C + B' \cdot C \cdot D' + A \cdot D + B' \cdot C' \cdot D'$$
- Draw the K-map of the complement of F, that is, F'.

A 4x4 Karnaugh map for the function F(A,B,C,D). The horizontal axis is labeled AB with values 00, 01, 11. The vertical axis is labeled CD with values 00, 01, 11, 10. The map contains red '1's in the following cells: (AB=00, CD=00), (AB=00, CD=01), (AB=01, CD=00), (AB=01, CD=01), (AB=01, CD=11), (AB=11, CD=00), (AB=11, CD=01), (AB=11, CD=11), and (AB=01, CD=10). The map is surrounded by lines labeled A, B, C, and D.

CD \ AB	00	01	11
00	1	1	
01	1	1	
11	1	1	
10		1	

From K-map,

F' =

Using DeMorgan's theorem,

F =

MORE EXAMPLES (6/6)

- Find the simplest POS expression for example #3:
 $F(A,B,C,D) = \Sigma m(2,8,10,15) + \Sigma d(0,1,3,7)$
- Draw the K-map of the complement of F, that is, F'.
 $F'(A,B,C,D) = \Sigma m(4,5,6,9,11,12,13,14) + \Sigma d(0,1,3,7)$

A 4x4 Karnaugh map for the function F(A,B,C,D). The vertical axis is labeled A (00, 01, 11, 10) and the horizontal axis is labeled B (00, 01, 11, 10). The map contains 1s at positions (A,B) = (00,01), (00,11), (01,11), (11,11), (10,01), and (10,11). There are also 1s at (00,00), (01,00), and (11,00) marked with red 'X's, indicating don't care conditions. The map is surrounded by labels A, B, C, and D with arrows indicating their respective axes.

AB \ CD	00	01	11	10
00	X	1	1	
01	X	1	1	1
11	X	X		1
10		1	1	

From K-map,

$F' =$

Using DeMorgan's theorem,

$F =$

THANK YOU
